

Design of a Precision Automatic Aiming System Based on K230 Visual Feedback and FOC Vector Control

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Abstract

To address the stringent requirements for dynamic response and positioning accuracy in simple automatic targeting systems, this paper presents a precision automatic targeting system based on an STM32F4 microcontroller and a K230 peripheral vision processor, overcoming limitations inherent in traditional servo control such as significant mechanical backlash, motor vibration at low speeds, and insufficient response bandwidth. The visual perception module, centered around the K230, employs a lightweight recognition algorithm utilizing geometric features and adaptive thresholds to achieve high-frame-rate target tracking exceeding 50 frames per second. The motion control module implements FOC control via the STM32F4, employing a cascaded control architecture and optimized PD strategy to eliminate motor torque pulsation at low speeds and suppress mechanical overshoot; the outer loop provides real-time angular correction through UART feedback. Tests demonstrate excellent system stability with a static targeting error of ≤ 1.5 cm, rapid dynamic tracking response, and smooth operation, fully meeting the demands for high precision and fast dynamic performance.

Keywords

Automatic Targeting System; Vision; FOC Vector Control; STM32

1. Introduction

With the rapid advancement of modern industrial automation and defense technologies, automatic targeting and precision tracking systems have become increasingly vital in areas such as target engagement, security surveillance, and laser guidance. These systems demand actuators with high dynamic response, exceptional positioning accuracy, and strong adaptability under complex lighting and mechanical conditions. Traditional simple pan-tilt designs predominantly employ stepper motors or PWM servos as motion execution units. However, in

applications requiring millisecond-level response times and sub-centimeter positioning accuracy, conventional solutions often face the trade-off between speed and precision: stepper motors exhibit low-speed vibration, while servos suffer from significant dead zones. Meanwhile, brushless DC motors have emerged as the preferred choice for high-end servo control due to their advantages of high power density, excellent torque characteristics, extended lifespan, and absence of brush-related interference [1, 2]. Magnetic field orientation control technology achieves high-performance control of AC motors by decoupling torque and flux components, and has seen widespread adoption in robotic servo systems in recent years [3].

For motion control applications in simple self-aiming systems, common technical solutions primarily include stepper motors, PWM servos, and brush-mounted DC motors. However, these approaches exhibit significant limitations in high-performance scenarios. Traditional stepper motor systems typically employ open-loop control; while structurally simple, they often experience "step loss" during high-speed startup or stopping, along with noticeable low-frequency vibrations and noise at low speeds. Standard servo systems offer limited position closed-loop control capabilities but face constraints due to mechanical clearance in internal gear sets and potentiometer linearity limitations, making precision below the centimeter level unattainable. Additionally, the low frequency of PWM drive signals restricts the system's dynamic response bandwidth. Brushed DC motors, despite offering broader speed regulation ranges, suffer from mechanical wear on brushes, electromagnetic interference caused by commutation sparks, and nonlinear frictional torque that severely impacts positioning accuracy at low speeds. To address the aforementioned limitations, this paper proposes a full closed-loop control scheme for brushless motors based on the STM32F4 microcontroller and Magnetic Field Orientation Control (FOC) technology. Compared to traditional six-step commutation methods, this approach employs Space Voltage Vector Modulation (SVPWM) to generate sine-wave voltages driving the motor, fundamentally eliminating torque pulsations and noise during low-speed operation [4]. The system utilizes an AS5600 magnetic encoder to obtain 12-bit high-precision position feedback data, combined with a position-based proportional-differential (PD) controller, achieving steady-state positioning accuracy below 0.1° . Furthermore, this study integrates machine vision with high-performance motor control by leveraging the edge computing capabilities of the K230 processor [5] to develop a target recognition algorithm based on geometric features and adaptive thresholds. To enhance control performance, the system adopts established visual servo theory [6,7] to establish a dual-closed-loop cascade control architecture comprising a "visual large loop" and a "motor small loop," effectively addressing cumulative errors caused solely by motor encoders. Additionally, a voltage-vector-based power-on self-calibration algorithm is developed to

automatically align the rotor magnetic field through fixed voltage vectors, resolving zero-point drift issues in incremental motion applications.

2. System Architecture Design

2.1. Distributed Hierarchical Hardware Architecture

The system employs a three-tier distributed decoupled architecture comprising perception, control, and drive modules, separating image processing from real-time motor motion control across separate processors. This design significantly reduces computational load on a single chip while ensuring synchronized and stable execution of both visual processing and motor servo control, as illustrated in Figure 1.

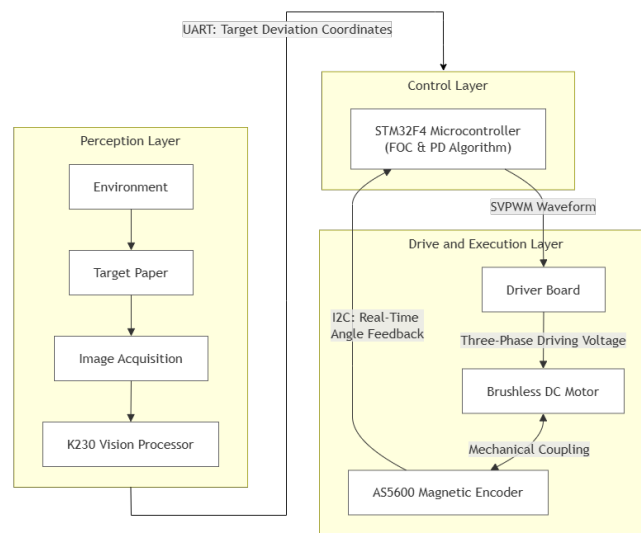


Figure 1. Overall Hardware Architecture of the System

The entire hardware platform consists of five components: the K230 visual perception unit, the STM32F407 motion control unit, the SimpleFOC three-phase power drive module, a brushless DC gimbal motor, and an AS5600 magnetic encoder. The visual perception layer uses the Canaan K230 edge processor as its core, which captures real-time scene grayscale images via the CSI image interface. It executes adaptive threshold segmentation and geometric feature filtering algorithms to calculate pixel offsets of target centers relative to the image optical axis, transmitting these deviation data to the host computer at a serial baud rate of 115,200 bps and a data update frequency of 50 Hz. The motion control layer employs the STM32F407 microcontroller [8], which receives pixel offset information from the serial interface, converts pixel coordinates into physical gimbal angles, and reads the 12-bit absolute rotor angle from the AS5600 magnetic encoder [9] via the I2C bus. This enables sequential execution of closed-loop position-based PD adjustment and SVPWM vector modulation algorithms. The actuation layer receives complementary PWM waveforms from the STM32's

advanced timer, which are amplified through the three-phase half-bridge inverter circuit in the SimpleFOC open-source hardware solution [10], driving the brushless motor to achieve precise closed-loop targeting of the laser-equipped gimbal. The overall signal flow is as follows: The target image is processed by the K230 to generate pixel deviations, which are transmitted via the UART serial port to the STM32 for conversion into target angle commands; the real-time mechanical angle of the motor rotor is fed back to the main controller through an I2C encoder, and after processing by the control algorithm, a three-phase PWM drive waveform is generated to ultimately control the gimbal for target alignment.

2.2. Visual Servo Dual Closed-Loop Cascade Control Strategy

To balance global target localization accuracy with local motor motion stability, the system employs a cascaded dual closed-loop control architecture comprising an outer visual position loop and an inner motor servo loop. The visual servo task is divided into a low-frequency position planning layer and a high-frequency motion execution layer, thereby mitigating control delays caused by image acquisition and algorithm computation, as shown in Figure 2.

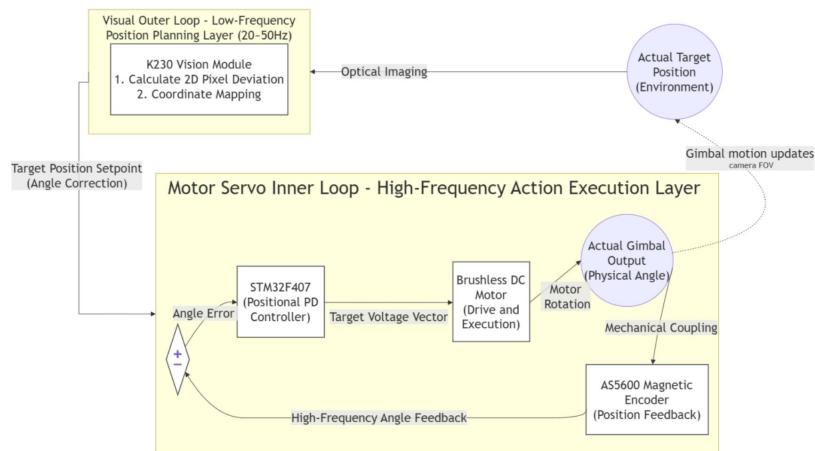


Figure 2. Dual Closed1-Loop Cascade Control Strategy for Visual Motors

The visual outer loop is independently managed by the K230 vision module, operating at a control frequency range of 20 – 50 Hz, with its primary function being to eliminate cumulative global positioning errors. The processor continuously calculates the two-dimensional pixel deviations between the target center and the image center point, performs coordinate mapping based on the fixed geometric relationship of the camera, converts pixel offsets into gimbal angle corrections, and dynamically updates the target position values in the motor inner loop to achieve real-time correction based on visual feedback. The motor servo inner loop consists of an STM32F407 microcontroller, an AS5600 magnetic encoder, and a brushless motor, capable of operating at control frequencies up to 1 – 10 kHz with rapid

command response and resistance to external mechanical disturbances. This inner loop continuously reads encoder angle feedback at high frequency, generates target drive voltage vectors via a PD controller, and employs differential components to provide virtual electronic damping, thereby suppressing overshoot and oscillations during gimbal startup/stop operations or approach to targets. In the dual-loop coordinated operation mode, the visual outer loop handles global position calibration while the motor inner loop executes high-speed, precise angular commands while maintaining smooth mechanical motion, effectively combining machine vision environmental perception with the high stiffness and fast-response characteristics of brushless motor servo systems.

3. Software Architecture Design

3.1. Software Layered Architecture

The overall software architecture adopts a layered design approach: visual processing tasks are handled by the K230 processor, while motor servo control tasks run on the STM32 microcontroller, with both components communicating via a UART serial interface. Developed using the HAL library, the STM32 microcontroller employs a front-end-back-end software architecture where the back-end module executes low-frequency sensor polling and system state machine management in a loop, whereas the front-end handles high-real-time tasks—including serial data reception interrupts and FOC timer control interrupts—to ensure timely responses for motor vector control and visual command interpretation.

3.2. Flow of the System Main Program

The overall system workflow is shown in Figure 3.

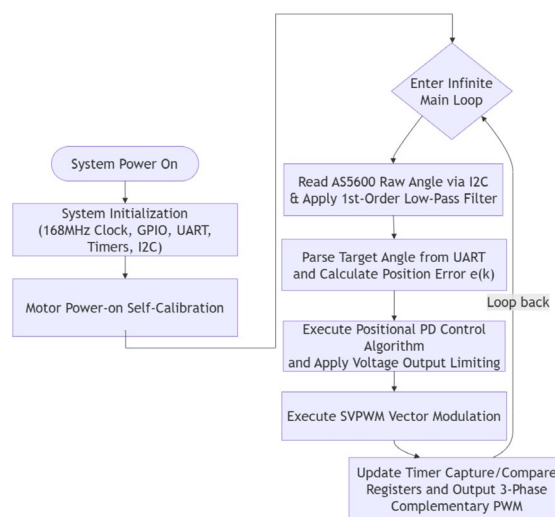


Figure 3. Overall workflow of the system's main program

Upon power-on, the system first completes full initialization: setting the STM32 clock frequency to 168 MHz and sequentially initializing all peripheral resources including GPIO pins, UART serial port, advanced timer, and I2C communication bus. Following initialization, the system executes a self-calibration procedure for the motor by applying a fixed-amplitude voltage vector to the stator to align the rotor magnetic field uniformly at zero position. The original encoder angle is then recorded as the zero offset value, compensating for inherent mechanical installation errors and addressing zero-point drift during incremental motion control. After calibration, the program enters an infinite loop performing multi-stage control tasks: reading AS5600's raw angle data via I2C hardware with first-order low-pass filtering to smooth noise; calculating positional error between actual and target angles using serial port data; executing position-based PD control algorithms to determine output voltage amplitudes with amplitude-clamping mechanisms; applying SVPWM vector modulation to generate PWM duty cycles for three-phase windings; updating the advanced timer's capture comparator register; and outputting complementary PWM waveforms to drive the motor through the power board.

3.3. Visual Recognition Algorithms

The system has developed a lightweight traditional machine vision recognition algorithm with high frame rate performance, as shown in Figure 4.

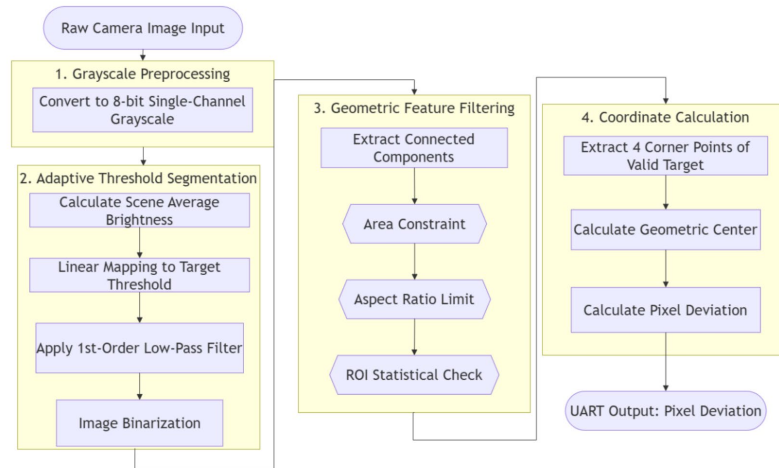


Figure 4. Visual Recognition Algorithm

The algorithm runs on the edge of the K230 chip, achieving a stable processing rate of over 50 frames per second. The specific algorithmic workflow is as follows:

The first step is the grayscale preprocessing stage. The system converts the raw images captured by the camera directly into a single-channel 8-bit grayscale format, reducing data volume by approximately 50% compared to traditional color formats and significantly improving image processing speed.

The next step involves the ambient light-adaptive threshold segmentation phase. To address threshold instability caused by fluctuating ambient lighting conditions, a dynamic exposure control algorithm based on first-order low-pass filters was developed, drawing upon classical image threshold segmentation theories [11,12]. The algorithm first calculates the scene average brightness in the image's central region and determines the target threshold using a linear mapping model. To ensure smoother threshold updates, the system employs a smoothing coefficient of 0.7 during time-domain processing, as illustrated in Formula 1:

$$\text{Thr}_{\text{curr}} = \alpha \times \text{Thr}_{\text{prev}} + (1 - \alpha) \times \text{Thr}_{\text{target}}. \quad (1)$$

Here, α represents the smoothing coefficient; this strategy effectively suppresses flickering in binary images caused by sudden changes in ambient light, ensuring consistent feature extraction.

The next phase involves geometric feature selection. The system employs a connected component extraction function to identify all regions in the image that meet specified criteria, and eliminates irrelevant interference targets based on geometric shapes and positional relationships. The selection process first applies area constraints to exclude noise points with excessively small areas. Subsequently, restrictions on aspect ratios eliminate striated interference objects that do not conform to target dimensions. Additionally, the system utilizes regional statistical feature verification: it defines a region around the center of the target's circumscribed rectangle as an interest region (ROI) and calculates its brightness mean value. Only when this mean exceeds a preset threshold is the target identified as a valid target, thereby significantly enhancing system stability.

Finally, the coordinate calculation stage is performed. The four corner points of the valid target are extracted, and their geometric center is calculated. Subsequently, the pixel deviation of this center relative to the optical axis center of the image is computed as a feedback error signal for the visual servo system, ensuring accurate target center detection under varying angles. The detection results are shown in Figure 5.

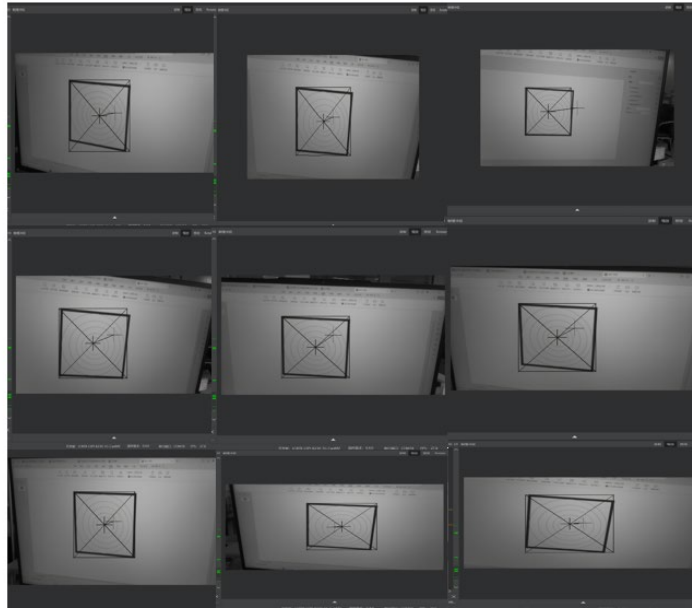


Figure 5. Identification and Detection Results

3.4. Visual Servo Communication and Coordinate Mapping

To establish a comprehensive visual feedback loop, the K230 must transmit calculated pixel deviations to the STM32 in real time. The system employs a non-blocking communication protocol based on ASCII character streams.

- Communication protocol design: The K230 device sends string commands in a specified format via UART, with the frame structure as follows:
 X-axis instruction: X + floating-point number string
 Y-axis instruction: Y + floating-point number string
- Lower-level machine parsing logic: The STM32 microcontroller activates the serial port receive interrupt mechanism via the HAL UART_RxCpltCallback function. During data transmission, each received byte is sequentially stored in a predefined buffer. Upon detecting a line break character in the command, the system immediately triggers its internal instruction parsing state machine. Specifically, if the first character is 'X' or 'x', the system calls the atof function to convert the subsequent string data into a floating-point number and updates the target angle (target_angle_A) for Motor A (yaw axis); if the first character is 'Y' or 'y', it updates the target angle (target_angle_B) for Motor B (pitch axis). This direct mapping mechanism enables the visual sensor to dynamically determine the gimbal's position in real time, achieving true "what you see is what you get" visual servo control performance.

3.5. Design of Closed-Loop Control for Motor Position Regulation

Given the physical characteristics of the pan-tilt system—its relatively constant load and low mechanical inertia—the system does not employ a complex three-loop

control structure but instead adopts a streamlined, highly efficient single-loop position-based proportional-differential (PD) control strategy. This approach ensures positioning accuracy while fully leveraging the control effects of the differential component to suppress mechanical vibration.

(1) Sensor Data Smoothing and Noise Suppression

During data acquisition, the AS5600 magnetic encoder is susceptible to electromagnetic interference and mechanical vibrations generated by motor operation, causing the directly acquired raw angle data ($\theta_{\text{raw_angle}}$) to contain high-frequency random noise. If this noisy data is fed directly into the PD controller, the noise is amplified through the differentiation term (D-term), leading to significant fluctuations in control signals and ultimately causing severe gimbal oscillation.

To address this issue, the system introduces a first-order low-pass filtering algorithm based on the weighted recurrence principle. This algorithm achieves dynamic smoothing of data by assigning distinct confidence weights to current sensor measurements and historical estimates. The specific filtering formula is as follows:

$$\theta_{\text{out}}(k) = \alpha \cdot \theta_{\text{meas}}(k) + (1 - \alpha) \cdot \theta_{\text{out}}(k - 1) \quad (2)$$

Here, $\theta_{\text{out}}(k)$ denotes the filtered output angle data at the current instant, $\theta_{\text{meas}}(k)$ represents the raw angle reading obtained by the sensor at that instant, and $\theta_{\text{out}}(k-1)$ indicates the filtered output value from the previous instant. The parameter α is the filtering coefficient, whose magnitude reflects the system's confidence level in the newly acquired measurement data.

During actual program code implementation, this system sets the α value to 0.1. This indicates that the system considers only 10% of the current sensor readings as valid "new information," while the remaining 90% should follow the "historical trend." This configuration differs from the threshold processing employed in the visual module; a smaller filtering coefficient is chosen here to effectively eliminate high-frequency noise that interferes with the derivative term, thereby achieving enhanced smoothing performance. From a physical perspective, this algorithm establishes a digital low-pass filter with a low cutoff frequency within the system by reducing sensitivity to abrupt data fluctuations. Consequently, the angle data curve input to the PD controller becomes smoother, significantly improving the operational stability of the differential control component and ultimately enabling flexible motion control of the gimbal mechanism.

(2) Control Law Design

The controller's input is the position error $e(k)$, and its output signal is the amplitude $U_{\text{out}}(k)$ of the control voltage vector. The discretized control formula is as follows:

First, the calculation error:

$$e(k) = \theta_{\text{target}} - \theta_{\text{filtered}} \quad (3)$$

Then calculate the controller output (the negative sign indicates reverse regulation):

$$U_{\text{out}}(k) = -[K_p \cdot e(k) + K_d \cdot (e(k) - e(k-1))] \quad (4)$$

Among the control parameters, the proportional term K_p primarily determines the system's responsiveness to errors. A larger value of K_p results in faster return to the target position; however, excessively high values may cause low-frequency oscillations. The derivative term K_d acts as an electronic damper: by detecting the rate of change of the error signal (i.e., angular velocity), it generates a counteracting torque when the motor rapidly approaches the target point, thereby preventing overshoot. This mechanism effectively suppresses motion overshoot and ensures the gimbal stops smoothly and without vibration at the predetermined target position.

(3) Safety Limit Mechanism

To prevent the control voltage calculated by PID from exceeding the tolerance limits of the drive board or motor, a constraint limiting function has been introduced:

$$U_{\text{final}} = \text{constrain}(U_{\text{out}}, -U_{\text{limit}}, +U_{\text{limit}}) \quad (5)$$

The system sets the voltage limit at $U_{\text{limit}} = 2.6 \text{ V}$, ensuring sufficient driving torque while eliminating risks of overcurrent and overheating.

3.6. Implementation of the FOC Vector Control Algorithm

The system employs voltage-based FOC (Field Oriented Control) technology to drive the brushless motor. Unlike traditional square-wave driving, FOC generates rotating voltage vectors in space to achieve continuous and smooth rotation of the rotor's magnetic field.

(1) Principle of Spatial Voltage Vector Generation

According to motor theory principles, to generate a voltage vector with amplitude U_{final} and electrical angle θ_{elec} on the stator's three-phase windings, the three-phase terminal voltages U_a , U_b , and U_c must follow a sine wave pattern.

$$\begin{cases} U_a = U_{\text{final}} \cdot \sin(\theta_{\text{elec}}) \\ U_b = U_{\text{final}} \cdot \sin\left(\theta_{\text{elec}} - \frac{2\pi}{3}\right) \\ U_c = U_{\text{final}} \cdot \sin\left(\theta_{\text{elec}} + \frac{2\pi}{3}\right) \end{cases} \quad (6)$$

(2) Software Implementation Process

The aforementioned algorithm is implemented in the `set_motor_pwm` function:

Electric angle calculation: The electric angle is calculated based on the mechanical angle θ_{mech} and the number of motor pole pairs P_n (this system has 7 pole pairs).

$$\theta_{\text{elec}} = \theta_{\text{mech}} \cdot P_n - \theta_{\text{offset}} + 3/2\pi \quad (7)$$

The value of $3/2\pi$ is used to align the magnetic field orientation coordinate system.

4. System Testing and Result Analysis

4.1. Setting Up the Testing Environment

The system testing environment is shown in Figure 6. The tracking area features a

100 cm × 100 cm white printed base plate, with 1.8 cm-wide black insulating tape applied around the perimeter as boundary markers. The target consists of A4-sized paper enclosed by a 1.8 cm-wide black rectangular frame, with a red concentric circle drawn at its center serving as the laser aiming reference point. The testing range covers 50 cm to 100 cm, fully aligning with the effective aiming range specified in the competition requirements. To verify the visual algorithm's adaptive threshold functionality, three lighting conditions were employed: standard indoor fluorescent lighting, strong lateral supplementary light interference, and a low-light dark environment with the main light source turned off. During testing, high-precision rulers and laser rangefinders were used to measure the linear deviation between the laser spot center and the target center, ensuring accurate error data measurement.

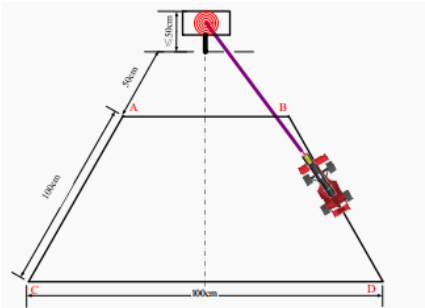


Figure 6. System Testing Environment Diagram

4.2. Static Aiming Accuracy Test

Static testing primarily evaluates the system's fixed-point aiming and positioning accuracy. The procedure involves randomly parking a vehicle equipped with a pan-tilt unit at various coordinates within the tracking area, activating the automatic aiming function, and measuring the linear offset distance between the laser spot and the target center once the spot stabilizes. A total of 10 datasets covering different distances and yaw angles were collected, with results presented in Table 1.

Table 1. Static Aiming Test Data.

Test Number	Distance to the target surface	Yaw Angle	total departure (cm)	Result Determination
1	50	0°	0.55	Qualified
2	60	+15°	1.13	Qualified
3	65	+20°	0.65	Qualified
4	70	+30°	0.47	Qualified
5	75	-15°	0.81	Qualified
6	80	-20°	0.60	Qualified
7	85	-30°	1.21	Qualified
8	90	+15°	0.87	Qualified
9	95	+20°	0.23	Qualified
10	100	0°	0.69	Qualified

Under all 10 test conditions, the system consistently identified targets and achieved automatic locking with a maximum static aiming deviation of 1.21 cm, yielding an average deviation of merely 0.64 cm across the ten datasets—significantly exceeding the competition requirement of ≤ 2 cm precision. At the maximum testing distance of 100 cm, laser spot jitter was minimal, eliminating dead-zone oscillations caused by traditional servo gear clearance issues. When switching between high-light, low-light, and standard illumination environments, the adaptive threshold vision algorithm reliably segmented black target boundaries, demonstrating excellent environmental robustness in geometric feature recognition.

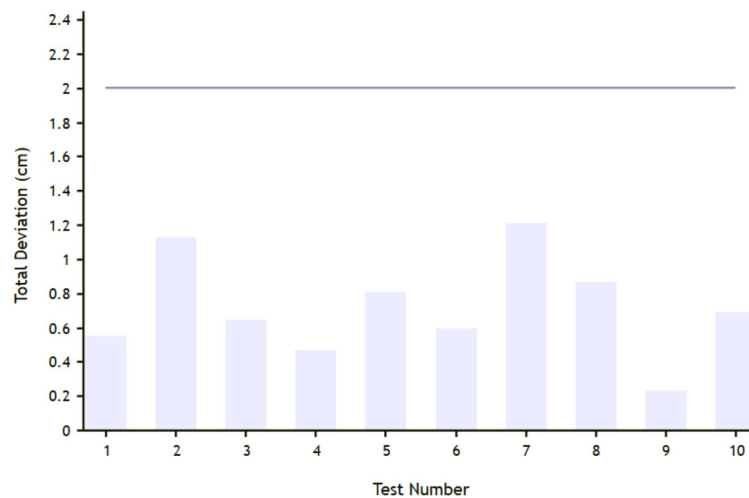


Figure 7. Distribution curve of static aiming deviation

4.3. Dynamic Tracking Performance Testing

Dynamic tracking testing evaluates system response speed and stability under

dynamic motion scenarios. The test configuration involves a target vertically mounted on the wall, with a vehicle equipped with an aiming gimbal moving linearly at a constant speed of 5 cm/s. The system continuously adjusts the gimbal's pitch and yaw angles to ensure the laser spot consistently follows the target's trajectory. Total system latency is determined by both image processing time and serial data transmission delay. Utilizing the K230's 50-frame-per-second image processing capability paired with a 115,200 bps high-speed serial interface, the total latency from target acquisition to motor-driven adjustment actions remains below 60 ms. During steady-state motion, minor lag errors persist in the laser spot; the PD controller's differential component ($K_d = 50$) provides continuous damping compensation to mitigate these effects. Upon sudden motion cessation, the differential component rapidly detects abrupt changes in error rate and applies reverse voltage vector control for braking damping, preventing significant overshoot or oscillation. This results in tracking performance far superior to traditional methods such as stepper open-loop control and single-proportion regulation.

5. Conclusion

This paper designs and implements a precision automatic aiming system that integrates K230 edge-side visual feedback with FOC vector control, completing the hardware platform setup, layered software algorithm development, and comprehensive performance testing under various operating conditions. The system comprehensively covers the entire process from image perception to servo control and power drive. At the motion control level, the STM32F4's floating-point unit enables voltage-based SVPWM-FOC brushless motor drive by replacing traditional square-wave driving with sine-wave voltage, thereby eliminating low-speed torque pulsation and operational noise at the fundamental level. Integrated with a 12-bit AS5600 magnetic encoder, it establishes a position-based PD closed-loop control; differential electronic damping suppresses gimbal overshoot, while an on-power self-calibration algorithm eliminates mechanical zero-point offset, achieving high-precision, low-jitter, and quiet gimbal servo control.

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